

Class #15 - LU Factorization

1. A *factorization* of a matrix $A \in \mathbf{R}^{m \times n}$ is an equation that expresses A as a _____ of two or more matrices.
2. Motivation: It is fairly common in industry or business to solve equations with the same coefficient matrix like:

$$Ax = b_1, \quad Ax = b_2, \quad \dots \quad Ax = b_p.$$

If A is invertible:

For general $A \in \mathbf{R}^{m \times n}$: We assume that A can be reduced to an echelon form *without row interchanges* (later, we will treat general case).

3. LU factorization of A :

Suppose $A \in \mathbf{R}^{m \times n}$ that can be reduced to an echelon form *without row interchanges*. Then,

$$A = LU$$

where L is an $m \times m$ lower triangular matrix with 1's on the diagonal and U is an $m \times n$ echelon form of A .

Note: The matrix L is _____ (even when A is not!)

Questions:

- (a) If L is a unit lower triangular (that is, L is an $m \times m$ lower triangular matrix with 1's on the diagonal), What will be on the diagonal of L^{-1} ?
- (b) If L and M are both unit lower triangular, then LM is also _____.

4. Why is this useful?

When $A = LU$, the equation $Ax = b_1$ becomes _____. Let $Ux = y$.

One can find x by solving the *pair* of equations:

- (i) _____, and (ii) _____.

LU requires $2n^3/3$ operations (flops). A^{-1} takes $2n^3$ operations.

Solving $Ly = b$ and $Ux = y$ takes $2n^2$ flops. Multiplying A^{-1} to b takes $2n^2$ flops (but roundoff error may occur.)

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5. Example: It is true that

$$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 2 & -5 & 1 & 0 \\ -3 & 8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -7 & -2 & 2 \\ 0 & -2 & -1 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Use this factorization to solve $Ax = b$ where $b = \begin{bmatrix} -9 \\ 5 \\ 7 \\ 11 \end{bmatrix}$.

$$[L \quad b] = \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ -1 & 1 & 0 & 0 & 5 \\ 2 & -5 & 1 & 0 & 7 \\ -3 & 8 & 3 & 1 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -9 \\ 0 & 1 & 0 & 0 & -4 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = [I \quad y],$$

then reduce

$$[U \quad y] \sim [I \quad \text{_____}].$$

This takes less number of operations than reducing $[A \quad b]$ to $[I \quad x]$.

6. LU factorization algorithm: Suppose A can be reduced to an echelon form U _____ row exchanges. Then A can be reduced to U with only _____. Then, there exist elementary matrices E_i such that E_i are _____ matrix with _____ on the diagonal.

$$E_p \cdots E_1 A = U$$

Then $L = \text{_____}$.

Note: Inverse of each E_i above is a _____, and L is _____.

7. Example: Find an LU factorization of

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}.$$

$$\sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & 3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & 3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & 3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix} = \underline{\hspace{2cm}}$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$